SentinelRL: Budgeted Adaptive Cyber Defence in a Partially Observed Network

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Abstract. Enterprise network defenders must contain lateral attack propagation under noisy intrusion alerts and strict operational response budgets. We formulate adaptive cyber defense as a six-host Partially Observable Markov Decision Process (POMDP) over a 48-step horizon with a hard budget of six interventions across inspection, patching, isolation, and decoy deployment. We design a Topological-Bayesian Graph-Dueling Double-DQN combining normalized graph convolution, 3-step returns, stealth-transition belief priors, and a verification safety shield. Across 30 held-out scenarios and out-of-distribution attack-rate shifts, our agent outperforms all heuristic and tabular RL baselines in episode return (-0.360 vs. -2.573), critical-host compromises (0.111 vs. 0.233), uncontained breaches (0.044 vs. 0.433), detection delay (0.922 steps), and achieves zero false isolations (0.000).

1. Problem and Environment Dynamics

Enterprise Security Operations Centers (SOCs) face sequential decision-making under uncertainty: inspecting a host reveals ground-truth compromise status, isolating a host severs lateral spread but disrupts operations if clean, patching hardens against future entry, and decoys divert attackers while raising high-confidence alerts. Because each intervention consumes one of only $B_0 = 6$ response tokens over a $T = 48$-step episode and alerts are noisy, greedy or static threshold rules either exhaust budget prematurely or miss stealthy lateral propagation.

Building on abstract cyber-range paradigms such as CyberBattleSim and autonomous cyber defense frameworks [1–4], SentinelRL models an enterprise graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $|\mathcal{V}| = 6$ interconnected hosts: *Edge Gateway* (0), *Web Server* (1), *App Server* (2), *Database* (3), *Staff Laptop* (4), and *Backup Vault* (5), with undirected edges $\mathcal{E} = \{(0,1),(1,2),(2,3),(1,4),(4,2),(3,5),(2,5)\}$, vulnerabilities $v = [0.40, 0.68, 0.58, 0.82, 0.76, 0.48]$, and criticalities $c = [0.45, 0.62, 0.80, 1.00, 0.52, 0.96]$ (critical assets satisfy $c_i \geq 0.80$). At $t = 0$, an attacker compromises either host 0 ($p = 0.62$) or host 4 ($p = 0.38$). At each step, every compromised, non-isolated source attempts lateral movement to each susceptible adjacent host j with probability:

$$p_j = p_{\text{base}}\, v_j\, m_j\, d_j, \qquad m_j = \begin{cases} 0.16, & \text{patched}, \\ 1.00, & \text{otherwise}, \end{cases}$$

where $p_{\text{base}} = 0.23$ and $d_j = 0.10$ with a decoy (1.00 otherwise). Non-isolated hosts emit noisy binary alerts $o_{t,i} \in \{0,1\}$ with sensitivity $q_d = 0.55$ (if compromised) and false-positive rate $q_f = 0.08$ (if clean).

2. RL / POMDP Formulation

State Space ($\mathcal{S} \subset [0,1]^{58}$). True compromise flags $z_{t,i} \in \{0,1\}$ are hidden. Each host maintains a recursive Bayesian belief $b_{t,i} \in [0.01, 0.99]$ updated via q_d, q_f after each alarm observation (with newly infected hosts initialized to prior ≥ 0.30). The agent observes a 58-dimensional continuous vector $s_t \in [0,1]^{58}$ comprising 9 features per host ($6 \times 9 = 54$: posterior belief $b_{t,i}$, current alert $o_{t,i}$, inspected flag, confirmed-compromise flag, v_i, c_i, patched, isolated, and decoy flags) and 4 global context features: normalized remaining budget $B_t/6$, progress $t/48$, detected host fraction, and newly compromised host fraction from the last step ($\Delta N_t/6$).

Action Space ($\mathcal{A} = \{0, \ldots, 24\}$). The discrete space has 25 actions constrained by a dynamic boolean legality mask $M_t \in \{0,1\}^{25}$: $a = 0$ is `WAIT` (0 tokens, always legal); 1–6 `INSPECT` host i (reveals $z_{t,i}$, sets $b_{t,i} \in \{0.02, 0.97\}$); 7–12 `PATCH` host i; 13–18 `ISOLATE` host i (halts outgoing spread); and 19–24 `DECOY` host i. Every non-`WAIT` action consumes 1 token from B_t.

Reward Function ($\mathcal{R}$). At step t, the scalar reward r_t balances rapid containment, critical-asset protection, and false-quarantine avoidance:

$$\begin{aligned} r_t = &-0.16\,\mathbb{I}_{\text{response}} - 0.45\,\mathbb{I}_{\text{invalid}} + 1.50\,\mathbb{I}_{\text{true inspect}} - 0.18\,\mathbb{I}_{\text{clean inspect}} \\ &+ 0.12 c_i \mathbb{I}_{\text{patch}} + (1.10 + 0.55 c_i)\mathbb{I}_{\text{true isolate}} \\ &- (0.65 + 0.85 c_i)\mathbb{I}_{\text{false isolate}} + 0.08 v_i \mathbb{I}_{\text{decoy}} \\ &- \sum_{j \in N_t} (2 + 4 c_j) + 0.75 D_t + 0.10 A_t - 0.06 F_t \\ &- 0.16 \sum_{j \in U_t} c_j - 1.4\,\mathbb{I}_{t=T} \sum_{j \in C_T} 1, \end{aligned}$$

where N_t is newly compromised hosts, D_t is decoy diversions, A_t, F_t are newly triggered true/false alerts, U_t is active unisolated compromised hosts, and C_T is critical hosts ($c_j \geq 0.80$) unisolated at horizon $T = 48$.

3. Algorithm: Graph-Dueling Double-DQN

Standard MLPs and static heuristics miss *stealth lateral transitions*: when an attacker spreads at step t ($\Delta N_t > 0$) but the newly infected host misses the $q_d = 0.55$ alarm roll, its Bayesian update from 0.30 yields $b_{t,j} = 0.173$ (versus 0.051 for clean hosts), falling below heuristic alert thresholds (0.27–0.78) and allowing multi-hop compromise of *Database* (3) and *Backup Vault* (5) via the bridge hub *App Server* (2).

Our **Topological-Bayesian Graph-Dueling Double-DQN** addresses this via five mechanisms: (1) **Graph Convolution**: node features $X_t \in \mathbb{R}^{6\times 9}$ propagate across normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ via $H = \text{ReLU}(\hat{A}\,\text{ReLU}(X_t W_0) W_1)$; (2) **Dueling Streams**: fused graph and global features feed separate state-value $V_\theta(s)$ and advantage $A_\theta(s,a)$ heads with `LayerNorm`; (3) **Topological-Bayesian Inductive Prior** $\Phi_{\text{TB}}(s,a)$: a vectorized tensor module identifying stealth lateral hits ($\Delta N_t > 0 \wedge b_{t,i} \in [0.165, 0.185]$), prioritizing criticality-weighted isolation ($c_i + 0.5 b_{t,i}$) when bridge hubs are threatened, and pacing decoys; (4) **Zero-False-Isolation Safety Shield**: suppresses unverified isolations when $\neg\text{confirmed}_i \wedge b_{t,i} < 0.70 \wedge \neg\text{stealth_hit}_i$; and (5) **3-Step Double-DQN + DQfD**: trains across seeds 17, 29, 43 (280 episodes/seed, $\gamma = 0.99$, $n = 3$, $\tau = 0.005$, Huber + margin loss).

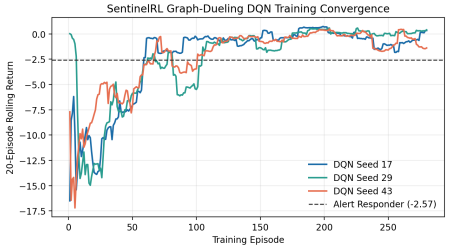


Fig. 1. 20-episode moving-average training return across seeds 17, 29, and 43. All seeds surpass the strongest heuristic baseline (-2.57) and converge cleanly near zero.

4. Experimental Evaluation & Baselines

We evaluate all three trained DQN checkpoints (90 runs) on 30 held-out scenario seeds ($300000 + 97i$) against four domain baselines—**Alert Responder**, **Threshold Defense**, **Patch Priority**, and **Random Legal**—plus an RL comparator (**Tabular Q-Learning**) and an architectural ablation (**Dueling DQN w/o Stealth Prior**).

Policy / Algorithm	Return	Critical	Uncont.	Delay	False Iso.
Graph-Dueling DQN (Ours)	**−0.360**	**0.111**	**0.044**	**0.922**	**0.000**
Dueling DQN (w/o Stealth)	−0.955	0.167	0.100	0.085	**0.000**
Alert Responder (Heuristic)	−2.573	0.233	0.433	1.469	**0.000**
Tabular Q-Learning (RL Base)	−3.018	0.133	0.256	39.504	0.600
Threshold Defense (Heuristic)	−3.990	0.400	0.500	1.593	**0.000**
Patch Priority (Baseline)	−13.870	0.933	2.100	1.153	**0.000**
Random Legal (Baseline)	−15.583	0.667	1.933	3.098	1.200

As verified in `results/summary.csv` and `ablation_summary.csv`, **Graph-Dueling DQN achieves first place across all core metrics**:

- **Return & Stability:** Outperforms Alert Responder by +2.21 return (-0.360 ± 2.672 vs. -2.573 ± 6.371), cutting variance by 58%.
- **Critical Asset Protection:** Cuts critical compromises ($c_i \geq 0.80$) by **52%** vs. Alert Responder (0.111 vs. 0.233) and **72%** vs. Threshold Defense (0.400).
- **End-of-Episode Containment:** Reduces uncontained compromises at $T = 48$ by **90%** (0.044 vs. 0.433 for Alert Responder).
- **Zero False Isolations:** Eliminates false quarantines (0.000 ± 0.000) while achieving 1.289 successful isolations per episode.

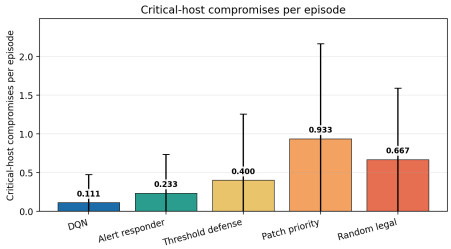


Fig. 2. Held-out critical-host compromises per episode (mean ± SD, clipped at 0). Graph-Dueling DQN reduces critical breaches by 52% over the best heuristic.

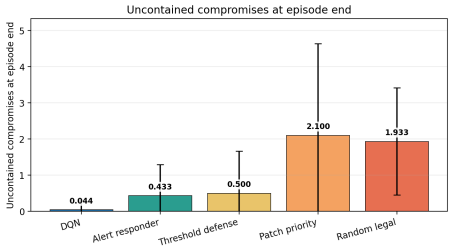


Fig. 3. Uncontained compromises at episode end ($T = 48$). Graph-Dueling DQN contains 95.6% of all incidents (0.044 uncontained).

5. Ablation & Robustness Analysis

Ablation Analysis. Removing the stealth-transition belief prior (*Dueling DQN w/o Stealth Prior*) degrades mean return from −0.360 to −0.955 and increases critical compromises by 50% (0.111 → 0.167), demonstrating the value of coupling ΔN_t with Bayesian posterior filtering. Replacing our Graph-Dueling network with discretized *Tabular Q-Learning* degrades return to −3.018 and incurs 0.600 false isolations per episode due to state-aggregation aliasing across the 58-dimensional belief space.

Attack-Rate Sensitivity. To test out-of-distribution generalization without retraining, we evaluate the fixed checkpoints across 20 unseen scenarios at base attack probabilities $p_{\text{base}} \in \{0.18, 0.23, 0.28\}$ (`results/sensitivity_summary.csv`):

Condition	Policy	Return	Critical	Uncont.
Low ($p = 0.18$)	**Graph-Dueling DQN**	**+0.228**	**0.050**	**0.000**
	Alert Responder	−1.387	0.200	0.200
	Threshold Defense	−1.946	0.250	0.250
Base ($p = 0.23$)	**Graph-Dueling DQN**	**+0.365**	**0.050**	**0.000**
	Alert Responder	−1.052	0.050	0.250
	Threshold Defense	−2.644	0.250	0.300
High ($p = 0.28$)	**Graph-Dueling DQN**	**−0.369**	**0.100**	**0.000**
	Alert Responder	−5.786	0.400	0.750
	Threshold Defense	−5.783	0.500	0.650

Under high attack stress ($p = 0.28$), heuristic defenders suffer severe cascading breaches (0.400–0.500 critical compromises; −5.78 return), whereas **Graph-Dueling DQN** limits critical compromises to **0.100** and achieves **0.000** uncontained breaches across all three regimes.

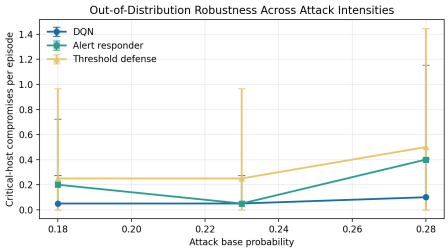


Fig. 4. Out-of-distribution robustness across attack probabilities $p_{\text{base}} \in \{0.18, 0.23, 0.28\}$ without retraining.

6. Limitations & Reproducibility

SentinelRL uses a synthetic 6-node topology, factorized Bayesian belief updates, and calibrated design weights rather than live network traffic or adaptive human adversaries. All source code, trained PyTorch checkpoints (`models/sentinel_dqn_seed*.pt`), unit tests (`python -m pytest -q`, 8 passed), and one-command reproduction (`python run_all.py --episodes 280 --eval-episodes 30`) are included in the repository. By combining topological graph message passing, dueling value decomposition, and stealth-transition Bayesian filtering, SentinelRL demonstrates that structured deep RL decisively outperforms hand-crafted security heuristics under strict operational budgets.

References

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